

Simran Verma

Embedded Systems & IoT Firmware Engineer • 11 Years Experience • MERIQ Index: 98% (Mastered)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: C/C++, FreeRTOS, ARM Cortex, PCB Debugging, Embedded Linux, I2C/SPI/UART

Verified Certification: Embedded Linux Developer Certificate

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Bosch* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core C/C++ service, reducing latency by 42%.
- Led a squad of 6 engineers delivering scalable microservices processing 21M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 35%.

Associate Embedded Systems & IoT Firmware Engineer — *STMicroelectronics* (2015 - 2023)

- Designed and maintained FreeRTOS data endpoints with 98% SLA uptime.
- Optimized database indexing and caching layers using ARM Cortex saving \$11k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [C/C++, FreeRTOS, ARM Cortex, PCB Debugging]

Engineered end-to-end production solution leveraging C/C++ and FreeRTOS with full test automation.

Low-Power BLE Sensor Node Firmware for Industrial Telemetry in FreeRTOS [ARM Cortex, PCB Debugging, Embedded Linux, I2C/SPI/UART]

Architected high-resilience system integrating ARM Cortex with 99.9% uptime.

EDUCATION & CREDENTIALS

B.S. in Cybersecurity and Information Assurance

IIT Bombay — 2011 - 2015 (Score: 3.92 / 4.0)